INCH-POUND

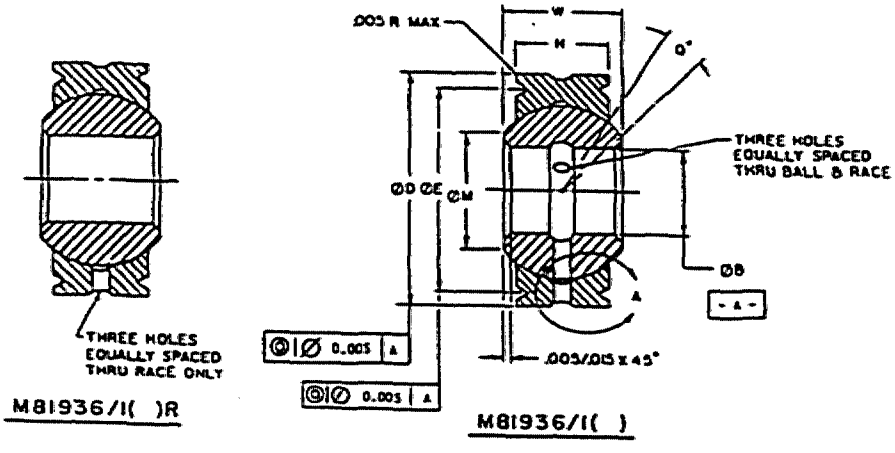
MIL-B-81936/1A
22 November 1993
SUPERSEDING
MIL-B-81936/1
28 SEPTEMBER 1973

MILITARY SPECIFICATION SHEET

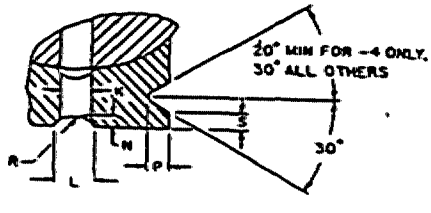
BEARINGS, PLAIN, SELF-ALIGNING, Be Cu BALL, CRES RACE, (WITH STAKING GROOVE) -65°F TO +350°F

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-81936



NOTE:
"D" & "E" CONCENTRIC WITH "B" WITHIN .005 TIR



DETAIL A

AMSC N/A

FSC 3120

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TABLE I. Dimensions.

DASH NUMBER		B	D	E	H	BALL DIA	K	L	M	N	P	Q*	R	S	W
LUBE GROOVES AND OIL HOLES IN RACE ONLY	LUBE GROOVES AND OIL HOLES IN BALL AND RACE	BORE +.0000 -.0005	OUTSIDE DIA +.0000 -.0005	+.000 -.010	RACE WIDTH +.000 -.005	+.001 -.005	OIL HOLE DIA THRU RACE AND BALL	GROOVE WIDTH ID & OD OF RACE & ID OF BALL	MIN	GROOVE DEPTH ID & OD OF RACE & ID OF BALL	+.000 -.015	MIN	GROOVE RADIUS ID & OD OF RACE & ID OF BALL	+.000 -.010	BALL WIDTH +.000 -.005
4R	4	.2500	.6562	.596	.250	.500	.032/.062	.042/.078	.357	.010/.015	.030	12	.030/.062	.020	.343
5R	5	.3125	.7500	.652	.281	.562	.042/.062	.042/.078	.413	.010/.015	.040	11	.030/.062	.030	.375
6R	6	.3750	.8125	.714	.312	.656	.042/.062	.042/.078	.509	.010/.015	.040	9	.030/.062	.030	.406
7R	7	.4375	.9062	.808	.343	.718	.052/.062	.065/.094	.563	.010/.015	.040	8	.060/.094	.030	.437
8R	8	.5000	1.0000	.878	.390	.813	.052/.062	.065/.094	.634	.010/.015	.060	8	.060/.094	.030	.500
9R	9	.5625	1.0937	.972	.437	.875	.052/.062	.065/.094	.464	.010/.015	.060	8	.060/.094	.030	.562
10R	10	.6250	1.1875	1.065	.500	.968	.062/.078	.073/.109	.732	.010/.015	.060	8	.070/.125	.030	.625
12R	12	.7500	1.4375	1.315	.593	1.187	.062/.078	.073/.109	.913	.010/.015	.060	8	.070/.125	.030	.750
13R	13	.8125	1.5625	1.440	.650	1.281	.062/.078	.073/.109	.984	.010/.015	.060	8	.070/.125	.030	.812
14R	14	.8750	1.6562	1.534	.703	1.375	.062/.078	.073/.109	1.054	.010/.015	.060	8	.070/.125	.030	.875
16R	16	1.0000	1.8750	1.753	.797	1.562	.078/.093	.082/.109	1.193	.010/.015	.060	8	.090/.125	.030	1.000
18R	18	1.1250	2.1250	2.003	.900	1.760	.078/.093	.082/.109	1.334	.010/.015	.060	8	.090/.125	.030	1.125
20R	20	1.2500	2.3125	2.190	1.000	1.937	.078/.093	.082/.109	1.473	.010/.015	.060	8	.090/.125	.030	1.250
22R	22	1.3750	2.5625	2.440	1.100	2.156	.078/.093	.082/.109	1.654	.010/.015	.060	8	.090/.125	.030	1.375
24R	24	1.5000	2.8125	2.690	1.200	2.344	.078/.093	.082/.109	1.794	.010/.015	.060	8	.090/.125	.030	1.500

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TABLE II. Axial and radial loads.

Dash No.	Radial static limit load LB	Axial static limit load LB	Peak radial load mode I LB	Peak radial load mode II LB
4	6330	1930	2570	5000
5	8460	2450	3520	6300
6	11400	3090	4570	8200
7	14800	3740	5750	9900
8	20400	4860	7500	12650
9	26700	6100	9500	15300
10	33100	8080	11750	19300
12	50000	11440	16900	28200
13	59000	13800	19800	33400
14	70300	16160	23000	38700
16	77700	20850	30000	49800
18	121500	26740	38000	63000
20	152000	33065	46900	77500
22	186000	40120	56900	95000
24	224000	47820	67500	112500

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REQUIREMENTS

1. MATERIAL

Ball - beryllium copper rod or bar per ASTM B194 or B196, TEMPER TH04.
Race - 17-4 PH per AMS 5643 condition H1150 per MIL-H-6875.

2. FINISH

Surface texture:
Spherical surface of ball and bore - 16 RHR.
Ball face and outer race O.D. - 32 RHR.
Outer race I.D. - 64 RHR.
All other surfaces - 125 RHR.

3. HARDNESS

Ball: Rc 37 min.
Race: Rc 28 to 36 before swaging.

4. LUBRICATION

Prepacked with MIL-G-81322.

5. TEMPERATURE RANGE

-65°F to +350°F.

6. DIMENSIONS

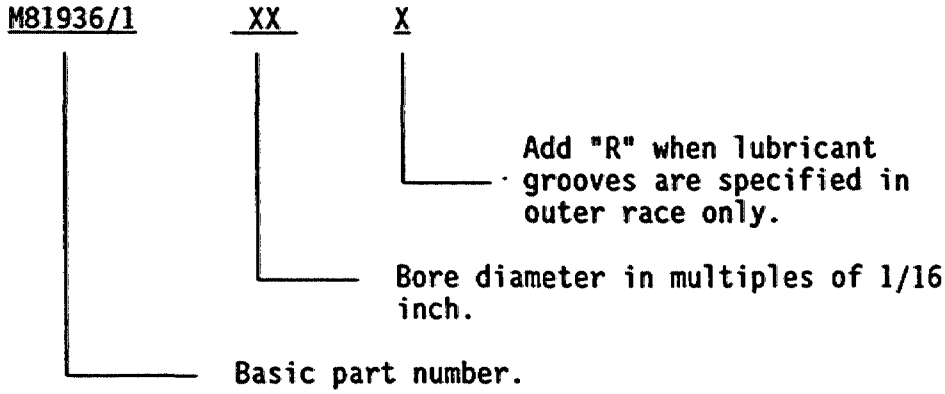
Dimensions in inches, unless otherwise specified. Groove dimensions on I.D. of race and grease hole dimensions through race are before bearing assembly but swaging shall not restrict grease flow.

Tolerances: Decimals ±.010
Angles ±1/2°

7. Break sharp edges and corners and remove all burrs and slivers.

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Example of part number.



Custodians:
Army - AV
Air Force - 99

Review activities:
Army - AR
Air Force - 84
DLA - IS

Preparing activity:
Navy - AS
(Project No. 3120-0718-01)

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